

Color-pattern diversity, *with and without history*

Demo 8 · pavo + phylogeny + measurement error

EEB 187 · Wednesday May 20, 2026

Today's question, in three parts

Are your family's color-pattern variables...

1. structured?
2. structured because of phylogeny?
3. structured robust to measurement noise?

Each part builds on the one before.

Last week — Frédéricich et al. 2026

the result this demo benchmarks against

$$K = 0.10$$

Chaetodontidae · multivariate Blomberg's K · 30 binary pattern motifs

Color patterns evolve fast.

Convergence and reversal across the tree.

Phylogeny barely predicts the next species' pattern.

Today's data (walkthrough)

Chaetodontidae mini-bundle

19 butterflyfish species — 5 genera, all matched to a pruned chronogram

1 of those 19 ships 5 images instead of 1 — the "error-species" for Part 3

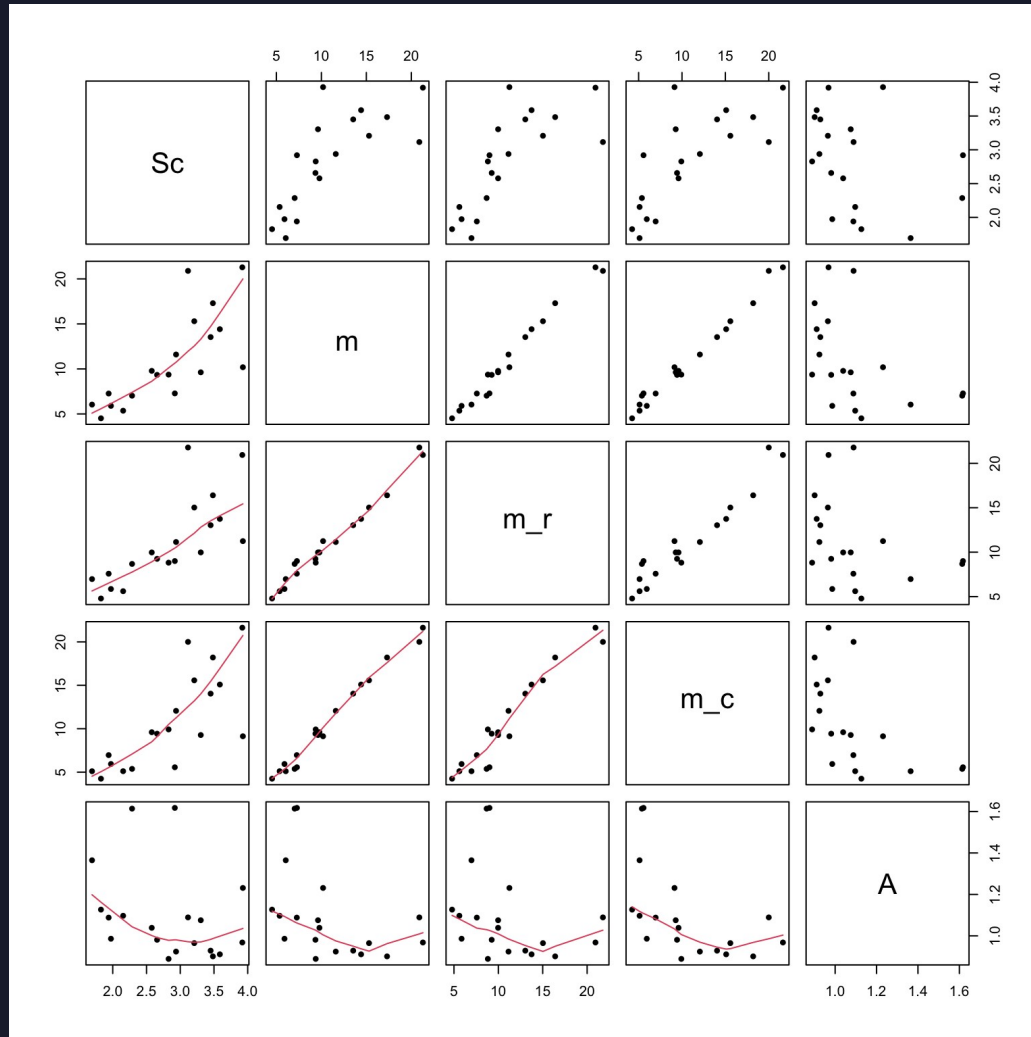
9 pavo pattern statistics per species

4 ecological traits from FishBase: trophic level, max depth, max length, vulnerability

Each group then re-runs the same workflow on their own family's mini-bundle.

Part 1 — describe what you've got

9 pattern stats per species · `pavo classify()` + `adjacent()`



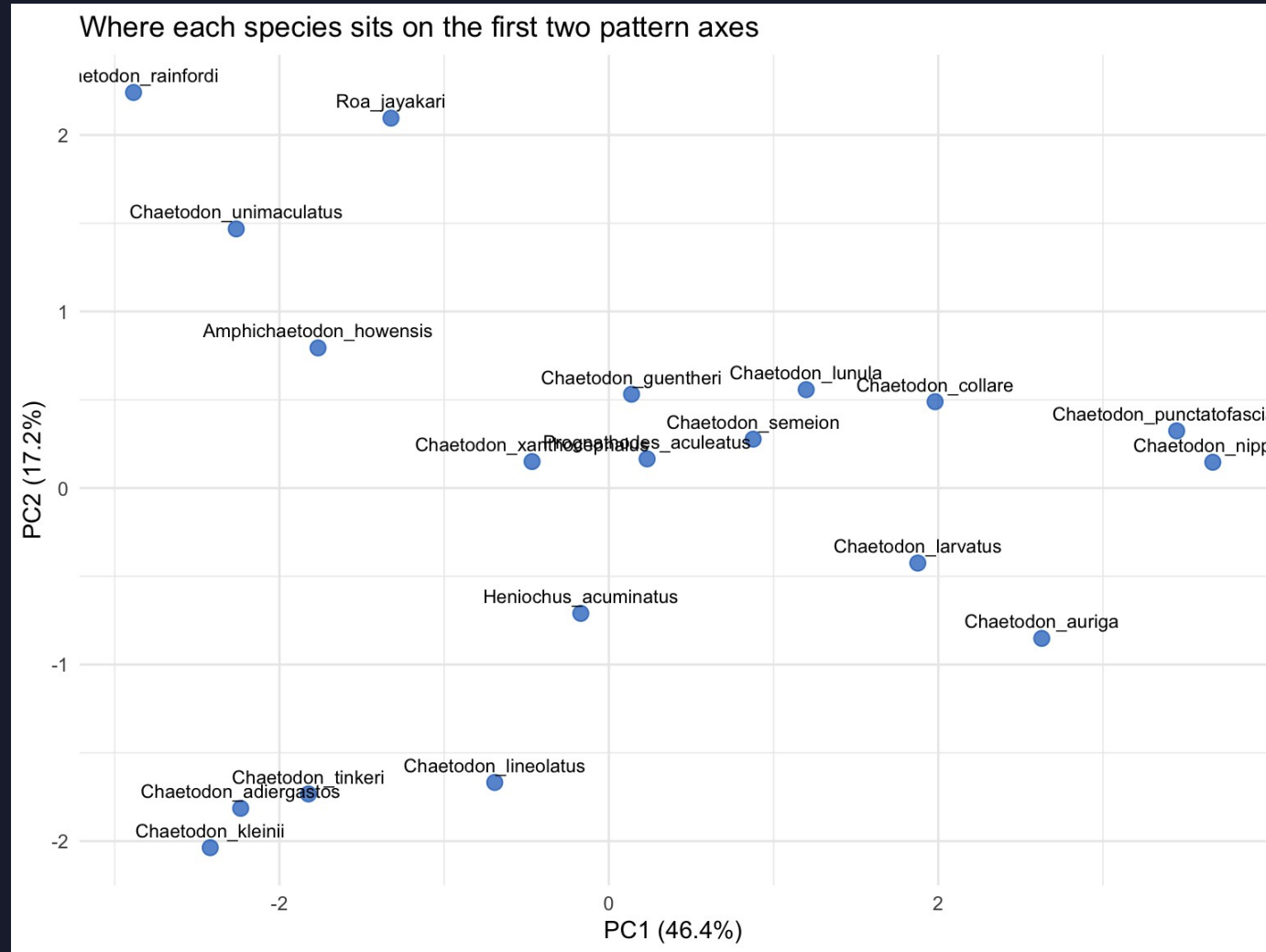
Diagonal: distribution

Off-diagonal: scatter +
smoother

Watch for tight scatter:
m, m_r, m_c all carry
the same information.

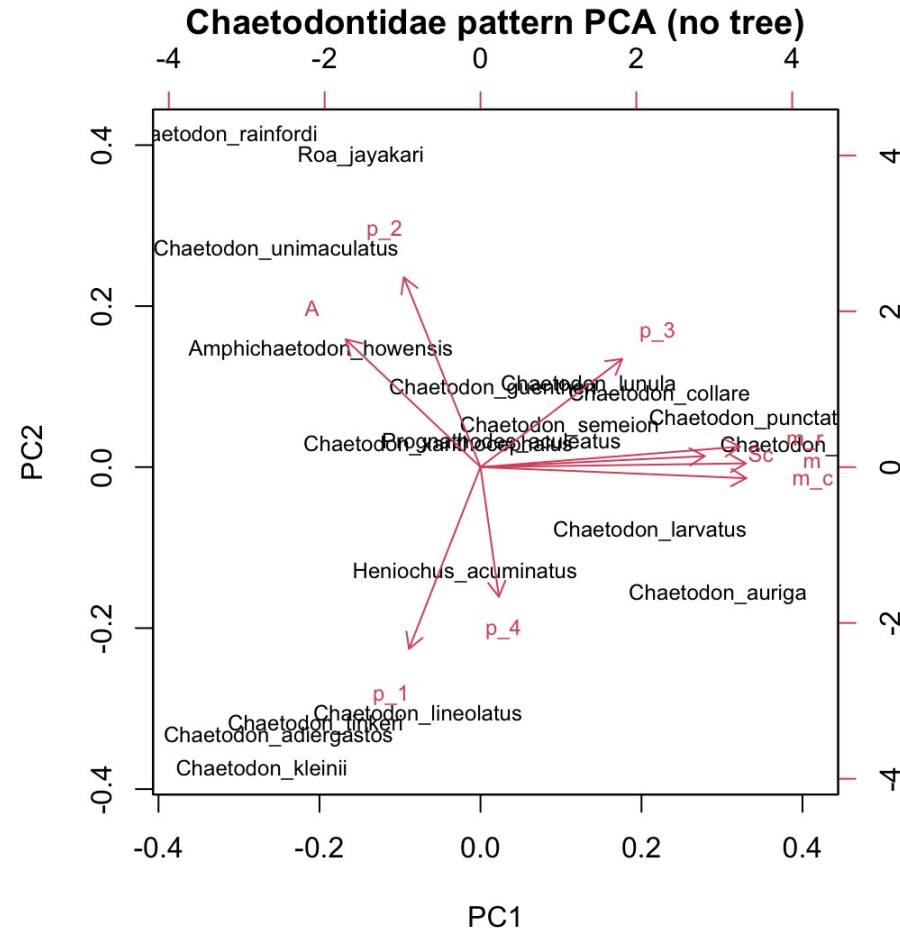
PC1 vs PC2 — where each species sits

PC1 $\approx$ 46 % · PC2 $\approx$ 17 % of pattern variation



What drives the two axes

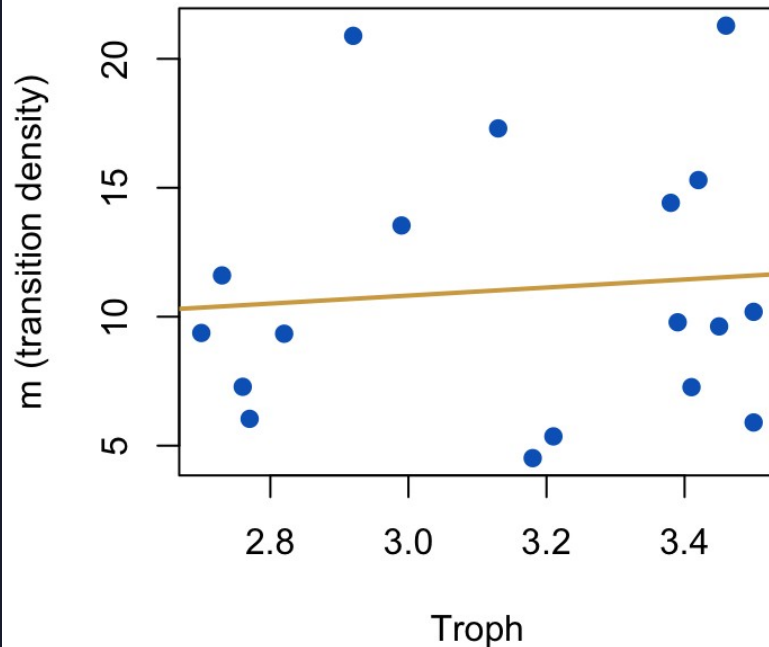
arrows = pavo variables · long arrow = high contribution



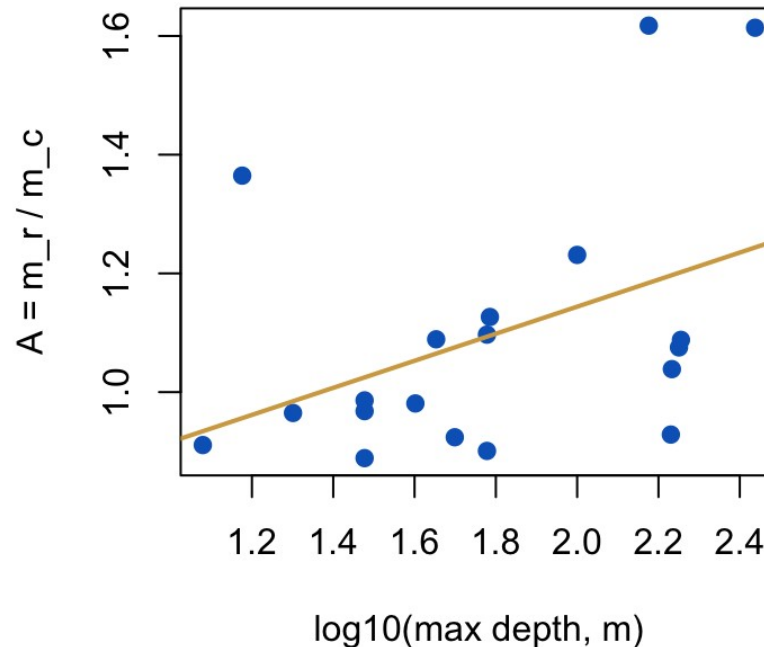
Does pattern track how the fish lives?

two named bivariate tests · ahistorical

Pattern complexity vs trophic level



Directional asymmetry vs depth



Trophic level →
complexity (m)

Depth →
asymmetry (A)

Part 2 will redo these
with the tree.

Part 2 — bring the tree back

Closely related species share trait values because they share genes.

Three corrections, in order:

phyl.pca — PCA reweighted by the tree's variance-covariance

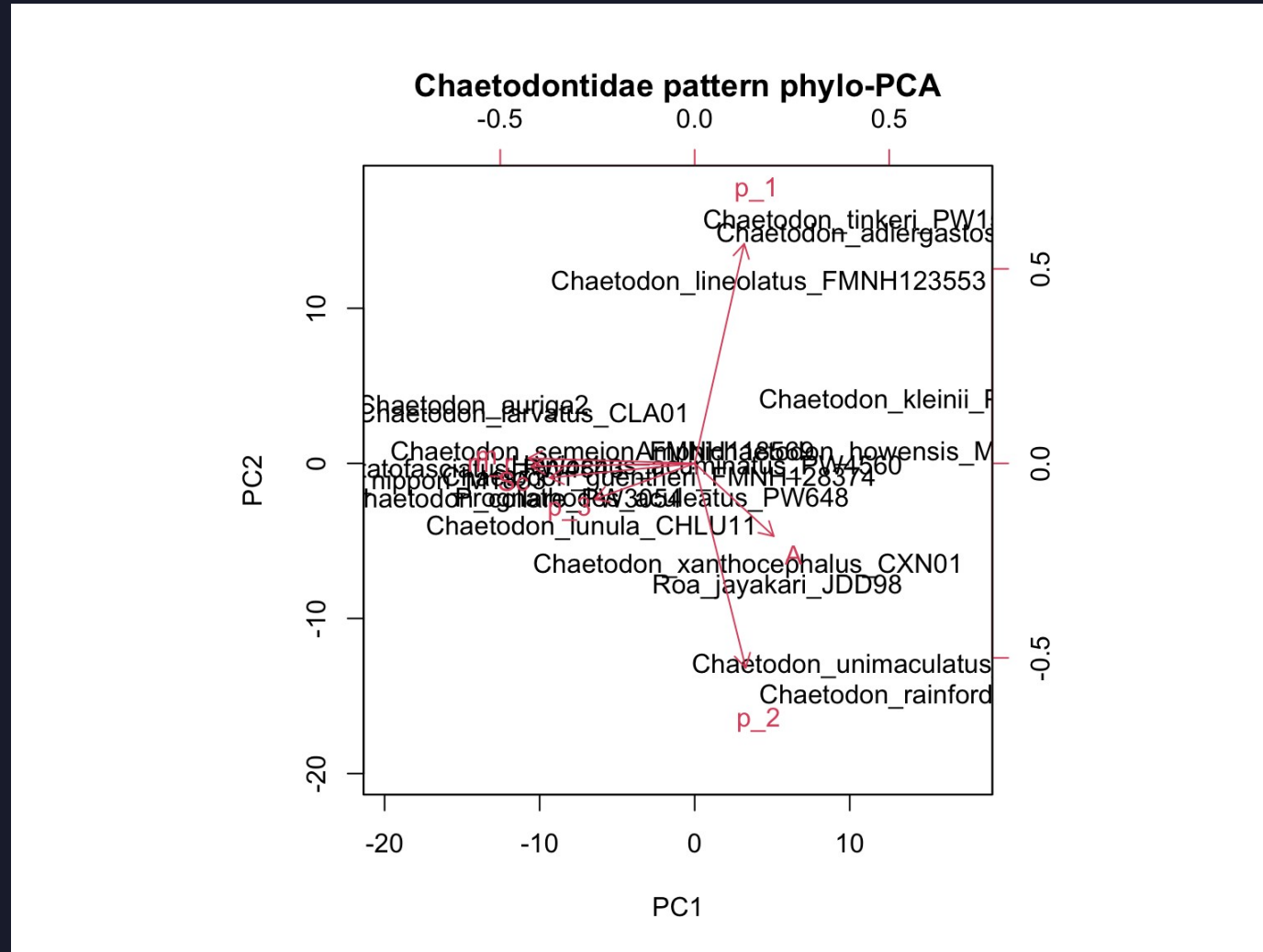
PGLS — bivariate regression where residuals can correlate via the tree

K and λ — phylogenetic signal, per variable

Same questions as Part 1. Different answer if phylogeny was driving things.

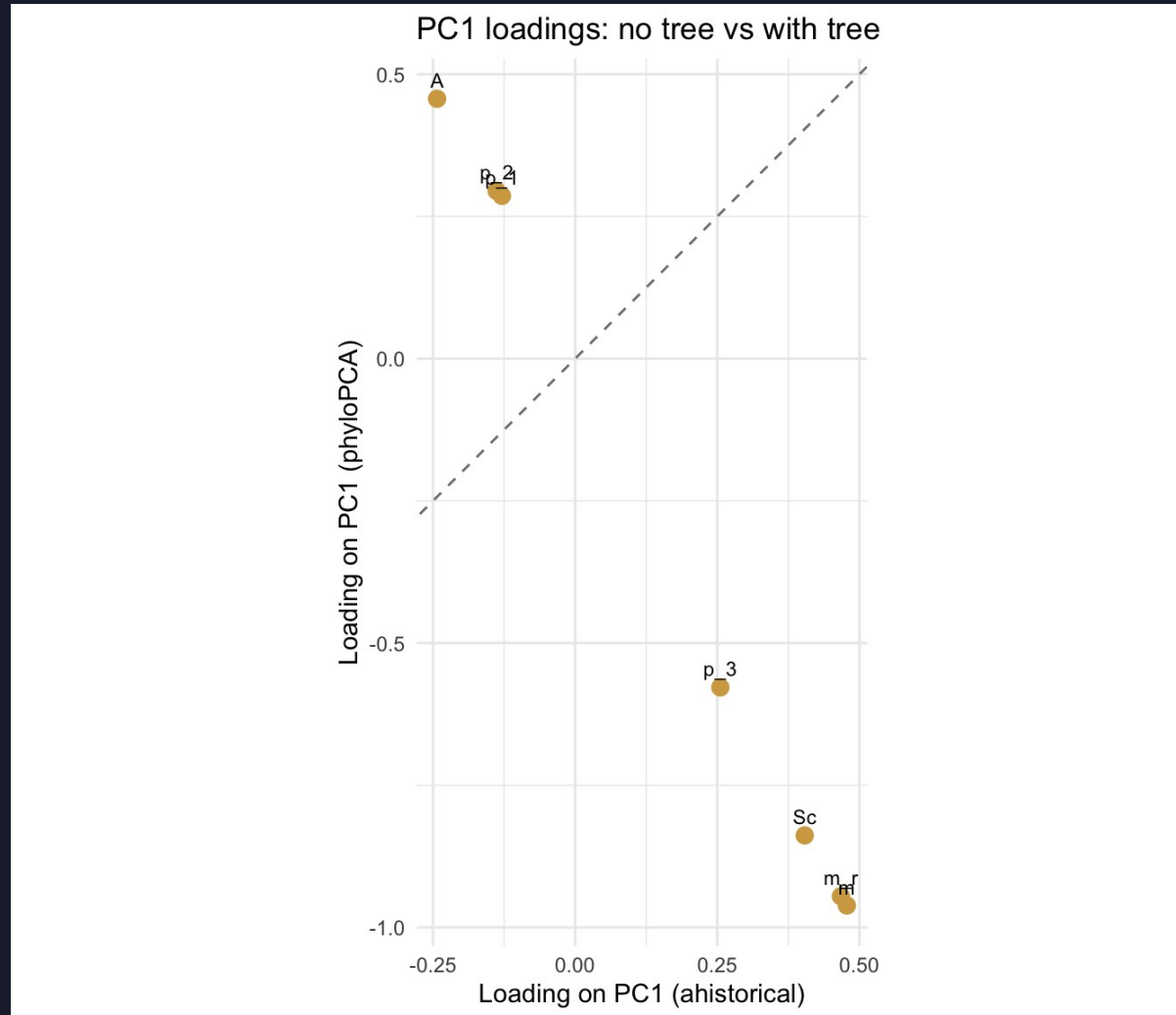
phyloPCA — same axes, different math

p_4 and m_c dropped because the trait covariance is rank-deficient



Did phylogeny change which variables matter?

PC1 loading: phyloPCA vs ahistorical

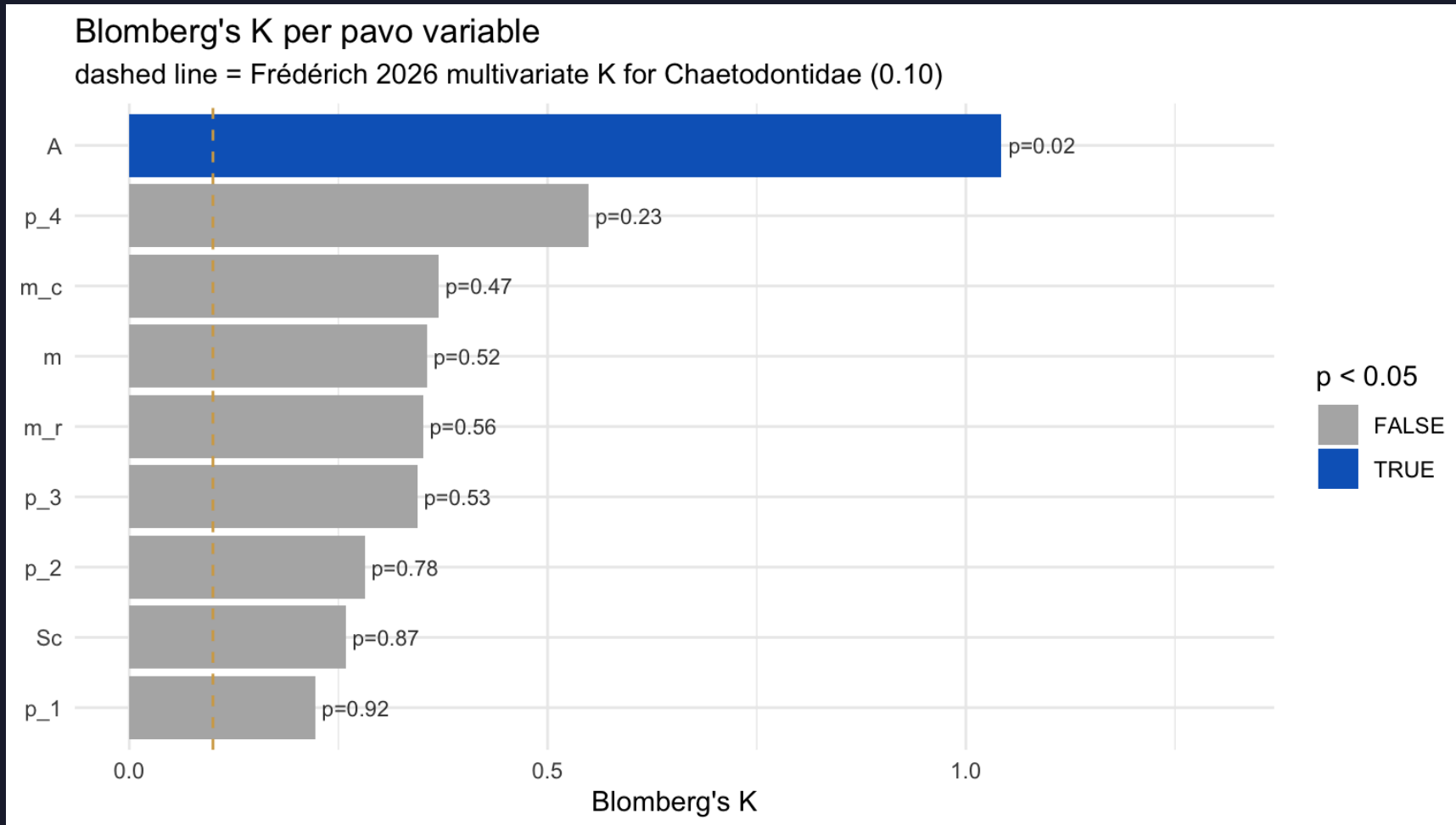


On the dashed line:
phylogeny
didn't change
that variable's
loading.

Far from it:
shared ancestry
was inflating it.

Blomberg's K per pavo variable

$K < 1$ = relatives more different than Brownian expectation

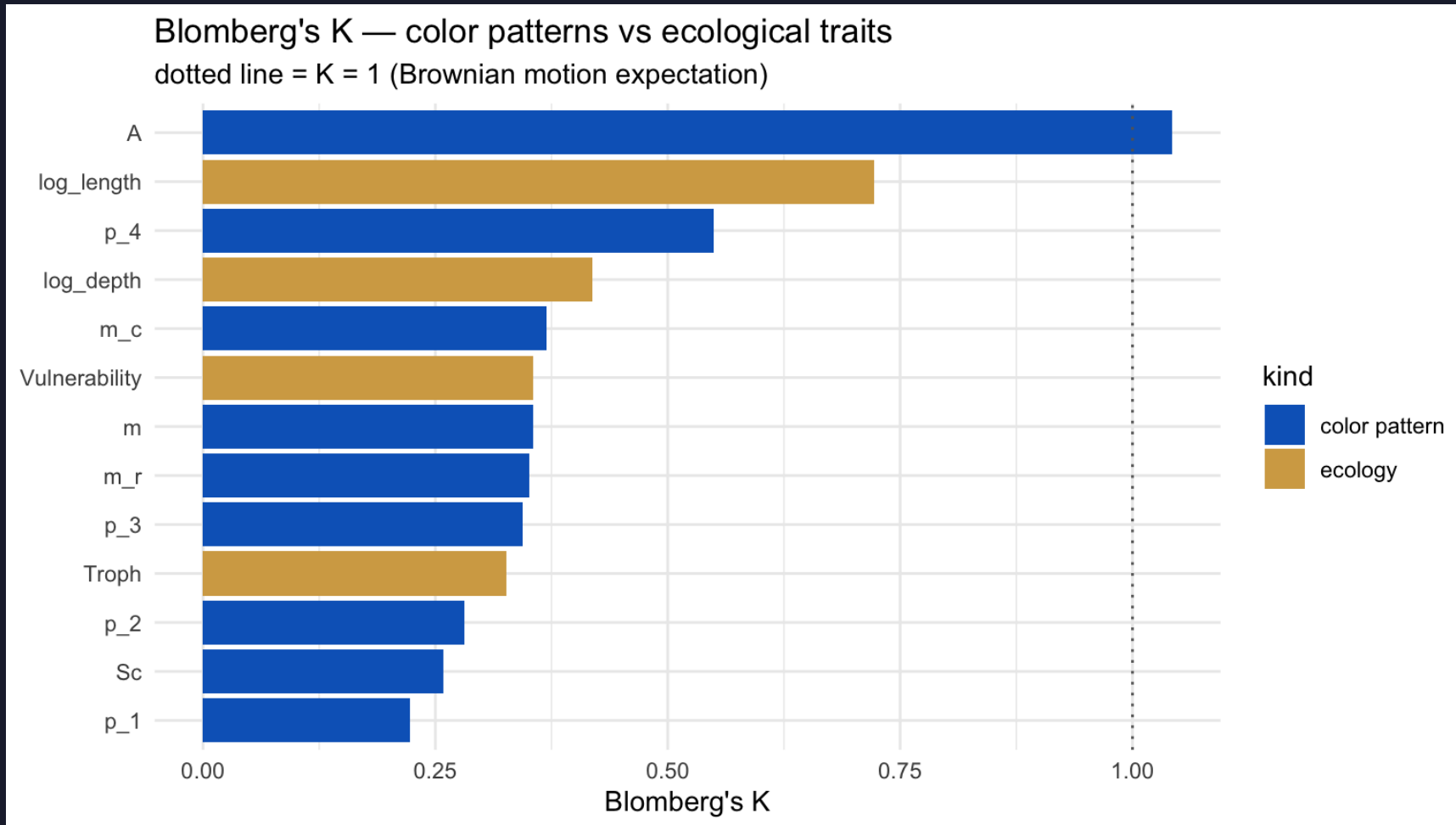


Blue bars:
 $p < 0.05$ —
real signal.

Dashed line:
Frédérich's
 $K = 0.10$ as
a landmark,
not a target.

Color signal vs ecology signal

where the demo lands



Dotted line:

$K = 1$, the

BM expectation.

Ecology bars

above color bars

= color evolves

faster than the

niche it tracks.

Part 3 — admit you got the numbers from a photograph

Pick one species. Photograph it five times.

Compute pavo on each photo.

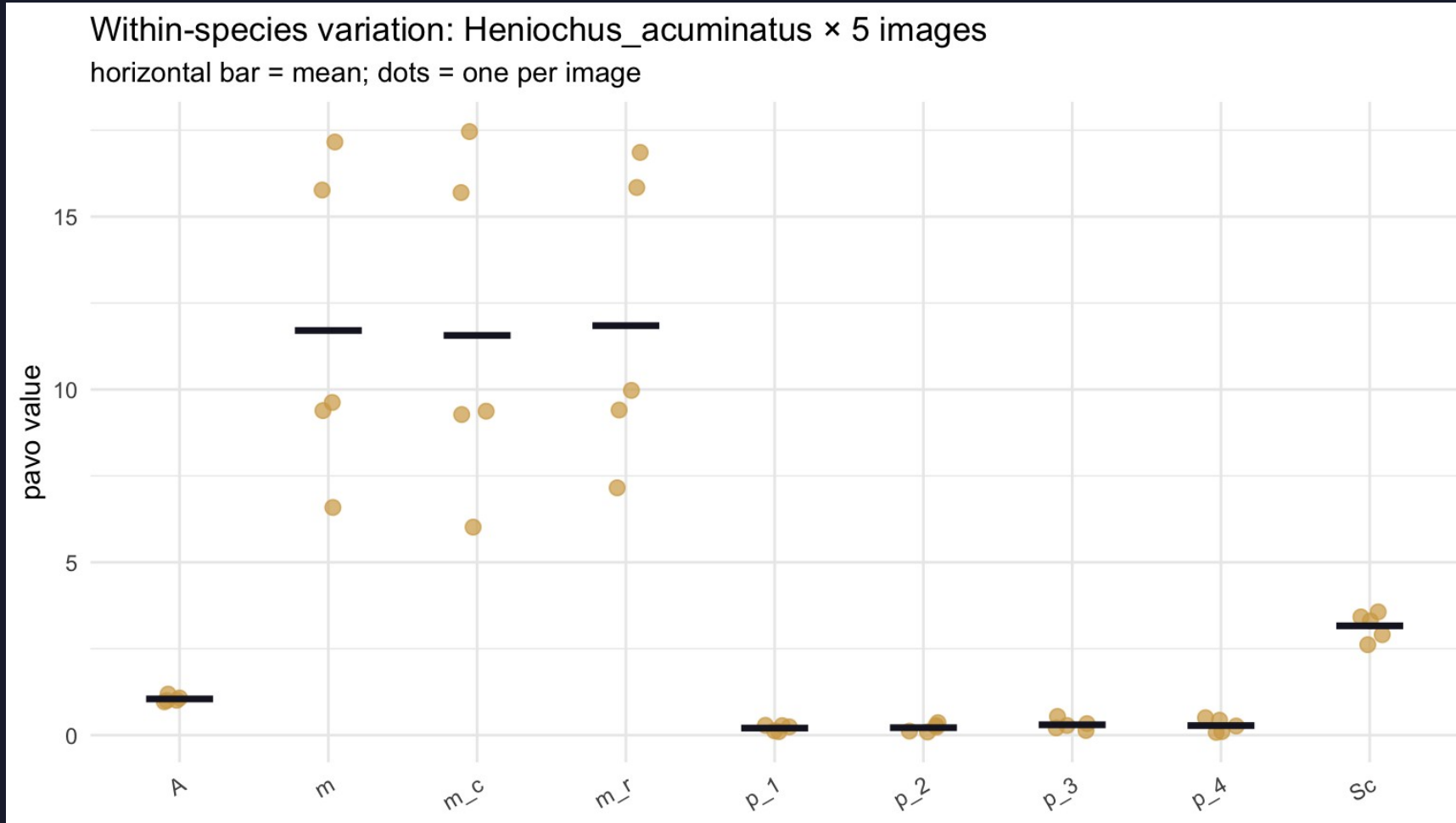
The spread is your measurement noise.

Now ask: how much of Part 2 survives noise of that size?

Monte Carlo: re-run Parts 1 & 2 a hundred times under that noise.

Within-species noise — Heniochus_acuminatus × 5 images

five dots per variable; bar = mean



Tight clusters

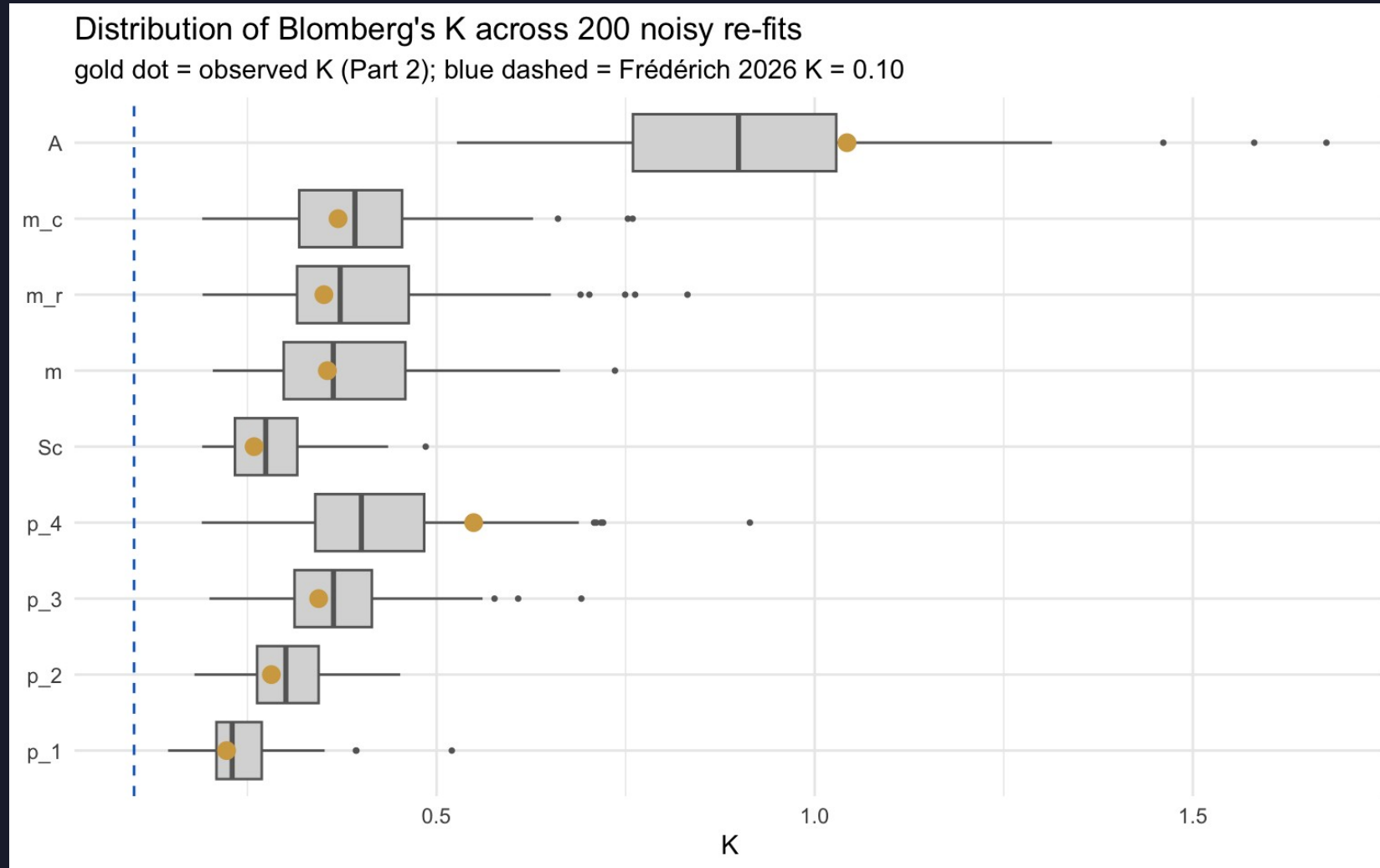
→ pavo is
reliable.

Wide spread

→ that variable's
Part 2 result
is fragile.

K under noise — 100 Monte Carlo re-fits

gold dot = observed K (Part 2)



Gold inside the box: robust.

Gold outside:
you got lucky —
a different photo
would have given
a different K.

What you turn in

BruinLearn · end of day Wednesday

1. Part 1 PCA scatter — your family
2. Part 2 K plot + one-line interpretation per variable
3. Part 3 Monte-Carlo CI on one statistic of your choosing
4. One paragraph (≤ 250 words): does color in your family match Frédéric, and which conclusions survive measurement error?

Tonight

prep page · ~20 minutes

1. Install R packages (one chunk on the prep page)
2. Download the Chaetodontidae bundle
3. Run the readiness check until it prints PASS
4. Confirm your group's mini-bundle login on FishView

michaelalfaro.github.io/eeb187-demos/demo8/demo8-prep.html

See you Wednesday.

Bring your laptop charged. We move fast.